# MANU KIRAN S

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## OBJECTIVE

Hands-on experience in Embedded Systems, IoT, and real-time hardware software integration. Electronics and Communication Engineering graduate seeking an entry-level Embedded Engineer role to design reliable and efficient embedded solutions.

## INTERNSHIP, APPRENTICESHIP & TRAINING

Vector India - Embedded Systems Training (Jan 31, 2026 - Present)

Bharat Electronics Limited (BEL) Trainee - HMC & RF Testing (July 2025 - Jan 2026) - Apprentice

- Tested RF modules and FPGA-based boards by verifying communication data, ADC values, and LO1/LO2 signals, ensuring all parameters were within specified limits.
- Performed PCB-level troubleshooting by measuring voltage, impedance, and RF signals across different frequencies, comparing output responses with expected specifications.
- Assisted in system calibration by checking signal accuracy, identifying errors, and ensuring the board was working within expected limits.
- Compared test results with reference values to identify faulty boards and confirm proper hardware performance.

Hindustan Aeronautics Limited (HAL) Helicopter Division RWRDC - (June 2025 to July 2025) -Internship

- Gained exposure to helicopter avionics, control systems, and module testing procedures.
- Observed real-time communication systems and system validation processes.
- Understood integration and reliability testing in aerospace embedded environments.

## TECHNICAL SKILLS

Programming: C, Embedded C, C++, Python
Embedded Platforms: Raspberry Pi, ESP32, Arduino Uno
Tools: Eagle PCB, MATLAB (basic), Fusion 360, Arduino IDE, ROS2 , Gazebo.
Core Areas: Embedded Systems, IoT, PCB Testing, FPGA Testing, Hardware Debugging, Verilog (Basic)

## EDUCATION

Presidency University, Bangalore
Bachelor of Engineering - CGPA: 6.93 - Electronics and Communication (2021 - 2025)

## CERTIFICATIONS

- Master ROS2 Basics and Become a Robot Operating System Developer | Step By Step | Robotics Programming | Python and C++ (Udemy)
- Advanced C (Udemy)
- Python (AI Hub)
- Eagle PCB (Coursera)
- MATLAB Onramp (MathWorks)

## ACADEMIC PROJECTS

1. **HIGHWAY SURVEILLANCE DRONE SYSTEM USING YOLOV8 (DJI TELLO + SERVER**)

• Developed AI-based drone system for real-time highway monitoring and vehicle detection.
• Integrated embedded hardware with computer vision for automated surveillance.
• Published research demonstrating intelligent transportation implementation.

2. Real-Time Object Detection for Autonomous Vehicles (Raspberry Pi 5 + Pi camera)

• Built an object detection system using YOLOv8 for self-driving navigation.
• Improved perception accuracy and system responsiveness.
• Used Python, OpenCV, and real-time processing techniques throughout development.

3. Land mine detection rover (ESP32+UItrasonic sensor + Metal detector sensor)

• Detects metallic land mines using a metal detector sensor and alerts the user when a potential mine is found.
• Uses an ultrasonic sensor for obstacle detection and autonomous navigation in hazardous environments

4. Smart Number Plate Detection (Raspberry Pi 4 + Pi camera)

• Built Raspberry Pi-based real-time number plate recognition using OpenCV.

5. Language Translator using OCR and Voice (Raspberry Pi 4 + Pi camera)

## FREELANCE PROJECTS

IoT Vending Machine with Online Payment (ESP32 + Razorpay)
• Developed an ESP32-based vending machine with Razorpay payment integration
• Ensured the product is dispensed only after successful payment confirmation
• Generates and displays a unique QR code for each transaction
• Optimized the system for low-cost and real-world deployment
• Hardware: ESP32, 1.8" TFT Display, optocoupler-based switching circuit, vending mechanism
• Software: C++, Flask, HTTP APIs, Razorpay

Smart Vending Machine with Dynamic QR Payment (Raspberry Pi 4 + Razorpay QR Code API)
• Developed a vending system using Raspberry Pi 4 that generates QR codes for each transaction
• Created a Flask-based backend to handle payments and control the system
• Enabled simple UPI payments and automatic product dispensing
• Hardware: Raspberry Pi , 1.8" TFT Display, optocoupler-based control circuit, vending mechanism
• Software: Python, Flask, Razorpay QR Code API

## PUBLICATION

• Published research on AI-powered Highway Surveillance Drone using YOLOv8.
• Demonstrated real-time embedded vision integration for intelligent traffic monitoring systems.